CHAPTER 5

# Trigonometric functions

S75, S76, S78, S79

**Objectives**

Use parity and periodicity; differentiate sine and cosine; solve equations and inequalities; study a periodic function.

## Course notes

### Circle and functions

On the unit circle, the point located by an angle $x$ in radians has coordinates $(\cos x, \sin x)$. For every real number $x$,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \leq \cos x, \sin x \leq 1.$$

Cosine is even: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. Both functions are $2\pi$-periodic.

| $x$ | $0$ | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ |
|---|---|---|---|---|---|
| $\cos x$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $0$ |
| $\sin x$ | $0$ | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | $1$ |

### Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi, \pi]$, cosine increases on $[-\pi, 0]$ from $-1$ to $1$, then decreases on $[0, \pi]$ from $1$ to $-1$. Sine decreases on $[-\pi, -\frac{\pi}{2}]$, increases on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2}, \pi]$.

**Justification.** Read the sign of sine and cosine on the circle, then apply the link between derivative and variations. Periodicity extends the curves; it does not remove the need to study an interval of sufficient length.

### Equations and inequalities

If $|a| > 1$, $\cos x = a$ and $\sin x = a$ have no real solution. For a known angle $\theta$,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

We then restrict the solutions to the required interval and eliminate duplicates.

**Examples on $[-\pi, \pi]$.** $\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \leq \frac{1}{2}$ gives $[-\pi, -\frac{\pi}{3}] \cup [\frac{\pi}{3}, \pi]$; $\sin x \geq \frac{1}{2}$ gives $[\frac{\pi}{6}, \frac{5\pi}{6}]$.

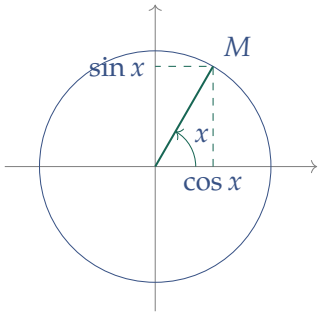
**A trigonometric composite**

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi, \pi]$, let $t = 2x \in [-2\pi, 2\pi]$, solve for $t$, then divide by 2. We obtain $x \in \{-\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\}$.

**Optimisation.** For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0, \pi]$, $f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function increases up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

**Double-angle identities used in the exercises.** For every real number $x$, $\sin(2x) = 2\sin x \cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their use in problems is not implicit.

**Symmetries of the curves.** The evenness of cosine corresponds to a symmetry about the vertical axis; the oddness of sine to a point symmetry about the origin. Periodicity extends each curve by translation by the vector $(2\pi, 0)$.




The coordinates of the point on the circle are $(\cos x, \sin x)$.

## Research activity

**Height of a point on a wheel — 35 min.**

A wheel turns at constant speed in a vertical plane. Its radius is 2 m and its centre is 3 m above the ground. A point on the wheel is at its lowest position at time $t = 0$. The wheel makes one revolution in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$, where $t$ is in seconds and $h(t)$ in metres.

1. Check that the model gives the stated initial height, a period of 8 s and the expected minimum and maximum heights.
2. Calculate $h'(t)$, then study the variations of $h$ on $[0; 8]$.
3. Determine all times $t \in [0; 8]$ for which the height is 4 m.
4. Determine the total time, during this revolution, for which the point is at least 4 m above the ground.

## Progressive exercises

**E1 · Practice.** Solve the equation $\sin x = -\frac{1}{2}$ in $[-\pi; \pi]$.

**E2 · Practice.** Solve the inequality $\cos x \geq \frac{\sqrt{2}}{2}$ in $[-\pi; \pi]$.

**E3 · Practice.** Define on $\mathbb{R}$ the functions $f(x) = \cos(3x - 1)$ and $g(x) = \sin^2 x$. Calculate their derivatives.

**E4 · Practice.** Consider $f(x) = 2 + \sin x$ on $[0; 2\pi]$. Calculate its derivative and draw up its table of variations.

**E5 · Practice.** Calculate $\displaystyle\int_0^{\pi} \sin x \, \mathrm{d}x$, then the mean value of the sine function on $[0; \pi]$.

## Textbook or past-paper work

**Barbazo, p. 166, exercise 19.** In S75: question 1, to read a symmetry and a period from a curve. At home: question 2, to prove the conjectures using identities valid for every real number.

## Homework

**Homework 5 — Splitting an area — 55 min, after S79.**

Consider the function $f$ defined on $[0; \frac{\pi}{2}]$ by $f(x) = \sin x \cos x$. Its curve is drawn in an orthonormal frame.

1. Calculate $f'(x)$ and determine the maximum of $f$ on this interval.
2. Solve on $[0; \frac{\pi}{2}]$ the inequality $f(x) \geq \frac{1}{4}$.
3. Calculate $\displaystyle\int_0^{\frac{\pi}{2}} f(x) \, \mathrm{d}x$, then the mean value of $f$ on $[0; \frac{\pi}{2}]$.
4. For $a \in [0; \frac{\pi}{2}]$, let $A(a) = \displaystyle\int_0^{a} f(x) \, \mathrm{d}x$. Determine $a$ so that the region under the curve between $0$ and $a$ has an area equal to half the total area under the curve between $0$ and $\frac{\pi}{2}$.